

LESSON GUIDE

Technology Plus

Unit 2: Programming

1.1.1 – Basics of Computing

Lesson Overview:

In this lesson, students will identify the four basic functions of computing and create a model of an example.

Students will:

- Differentiate the four basic operations of a computer.
- Create a model and label the four basic operations for a given computing task.

Guiding Question: What are the four basic operations we want all our computing devices to do?

Suggested Grade Levels: 8-12

Technology Needed: No

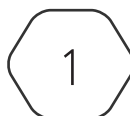
Approximate Time Required: 30-45 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 1.1 – Explain the basics of computing.
 - Input
 - Processing
 - Output
 - Storage

This content is based upon work supported by the US Department of Homeland Security's Cybersecurity & Infrastructure Security Agency under the Cybersecurity Education Training and Assistance Program (CETAP).



CYBER.ORG
THE ACADEMIC INITIATIVE OF THE CYBER INNOVATION CENTER

Copyright © 2025 Cyber Innovation Center
All Rights Reserved. Not for Distribution.

Lesson Background

Background Information:

Each of the four basic operations of computing will be covered more in depth in future objectives. This serves as an introduction and will help students begin to visualize the general assembly of a computer. The activity includes 10 problems – computing tasks. Groups will work together to identify the steps required to complete the tasks and label each step as one of the four basic operations. These same 10 problems will be used in 4.4.1 – Programming Organization – so encourage groups only to focus on the basic computing cycle rather than trying to write code/solutions to the problem.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 1.1.1 – Basics of Computing • Activity Slides 1.1.1 – Create a Computer
Materials per Group	• Activity Handout 1.1.1 – Create a Computer
Materials per Student	• Student Notes 1.1.1 – Basics of Computing • Student Handout 1.1.1 – Basics of Computing • Assessment 1.1.1 – Basics of Computing (optional)

Cyber Terms:

Term	Definition
input	user actions directed at the computer; the process of providing data or commands to a computer system so it can perform tasks or operations
process	computer actions performed to manipulate input data, execute instructions, and generate output
storage	mechanisms and devices used to save and retain data and programs for later use; saving output short-term (in system memory) or long-term (on hard drives)
output	result(s) of processing inputs; information or responses delivered to users or other systems

1.1.1 – Basics of Computing

Guiding Question: What are the four basic operations we want all our computing devices to do?

Lesson Launch (5-7 minutes)

- Distribute Student Handout 1.1.1 – Basics of Computing and encourage students to complete the handout as you go through today's lesson.
- The basic cycle of computing includes four main operations that repeat continuously: receiving input from the user, processing that input, using storage (long-term or short-term), and creating an output for the user.

Computing: That's So Basic! (10-15 minutes)

- Walk students through each of the four basics. Discuss a few examples for each, however, they will all be covered in greater detail later in the course.

Input

- Input is the process of providing data or commands to a computer system so it can perform tasks or operations.
- Methods:
 - Direct – user directly inputs data
 - Indirect – sensors or other devices gather and send data to computer
- Types of input devices:
 - Keyboard (**indirect** – you type commands and the computer translates key presses into computer language)
 - Mouse (**indirect** – you control the mouse and the computer reads a pointer on the screen)
 - Microphone (**indirect** – audio data is captured then processed)
 - Touchscreen (**direct** – you directly touch icons and buttons on screen)
 - Smart Speaker (**indirect** – audio data is captured then processed)
- Use the examples on slide 5 to check for understanding:
 - Video game controller (**indirect** – button presses and joystick moves are interpreted)
 - Stylus (**direct** – you control a pen-like input directly on screen)
 - Camera (**indirect** – visual data is captured then processed)
 - Fitness watch (**indirect** – sensors collect inputs like heart rate and step count)

A Grand Opening, Part 1

- We will use the example of a user wanting to open an application throughout the lesson to show the four basics parts of computing. Groups will then repeat a similar process in the Lesson Lab.
- A user wants to open an application. How do they let the computer know what they want the computer to do?
- Students may share different ways to input this request including clicking on an icon for an application (in a GUI) or typing a command (in a CLI). They may also share examples of using a touchscreen to directly click the application's icon, or a smart speaker to use a voice command.

Processing

- Processing refers to the actions performed by a computer to manipulate input data, execute instructions, and generate output, making it a fundamental aspect of computing.
- CPU – the brain – handles instructions with arithmetic, logic, control, and input/output (I/O) operations. Processor and CPU are used synonymously.
- Though this will be covered more thoroughly in Lesson 1.3.3 – Processing Speed, students may be interested to know that processing performance is measured by clock speed in gigahertz (GHz), core count, and instruction set.

A Grand Opening, Part 2

- Now that the user has provided input, clicking on an application icon, how does the computer process that request?
- Share that the computer will access its instructions for opening applications from the operating system stored in memory.

Storage

- Storage refers to the various mechanisms and devices used to save and retain data and programs for later use.
- Temporary/volatile/short-term
 - Primary storage
 - Random access memory (RAM) – fast
- Permanent/persistent/long-term
 - Secondary storage
 - Solid-state drive (SSD) – faster, more expensive
 - Hard disk drive (HDD) – slower, moving parts for recording
- Storage devices will be covered in depth in Lesson 2.3.1 – Storage Devices.

A Grand Opening, Part 3

- The processor has requested the application from long-term storage (HDD/SSD) and instructed the short-term memory (RAM) to run an instance of the application.

Output

- Output refers to the results generated by a computer after processing input data, delivering information or responses to users or other systems.
- A system can output information to the user through:
 - Displays/monitors
 - Speakers/headphones
 - Printers
 - Projectors
- Output provides a meaningful interaction between the user and the results of their computational tasks.

A Grand Opening, Part 4

- The computer received the user request to open an application, the processor worked with storage to access and run the application. How does the computer communicate back to the user that the application is open?
- Slide 12 is animated to show that the user receives a visual representation of the requested application on the display (output device).

Activity: Create a Computer (15-20 minutes)

- Divide students into groups of 2-4.
- Decide whether you want to assign a problem to each group or allow them to create their own.
- An example problem would look like this: “If a simple program asks the user to provide an integer, then gives back the result of doubling that integer, identify the four operations.”
 - Input – the user providing an integer
 - Storage – storing the user input as an integer
 - Processing – multiplying the integer by two
 - Output – displaying the doubled number to the user
- Consider displaying the group examples around the room for other groups to view. Then ask them to label the activity in terms of computing operations:
 - Input – teacher gives groups a problem/task
 - Storage – groups write down (store) instructions
 - Processing – groups create and label the computing process
 - Output – groups display their labeled problem

Putting It All Together (3-5 minutes)

- What are the four basic operations of a computer?

Input, Storage, Processing, Output

- What are two examples of how users provide input to a computer?

Keyboard, Mouse, Touchscreen, Microphone

- If a program calculates users' average daily heartrate, what are possible examples of the four basic operations taking place?

Input – user enters heartrate every few hours throughout the day or wears a heartrate monitor that sends the heartrate continuously to the program

Storage – stores each entered heartrate as an integer variable

Processing – adds all inputs and divides by total number of inputs for the day

Output – displays calculated average